

Motor Control Board

Variant: DRAFT

2026-08-10

Rev + (Unreleased)

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TOP VIEW

kibot_Mimage_png_3d_viewer_angled_top

BOTTOM VIEW

kibot_Mimage_png_3d_viewer_angled_bottom

DESIGN CONSIDERATIONS

DESIGN NOTE:
Example text for
informational design
notes.

DESIGN NOTE:
Example text for
debug notes.

DESIGN NOTE:
Example text for
cautionary design
notes.

DESIGN NOTE:
Example text for
critical design
notes.

LAYOUT NOTE:
Example text for
critical layout
guidelines.

NOTES

Comment

Not fitted components are marked as **X**

DRAFT - Very early stage of schematic, ignore details.

PRELIMINARY - Close to final schematic.

CHECKED - There shouldn't be any mistakes. Contact the engineer if you find any.

RELEASED - A board with this schematic has been sent to production.

Block Diagram

File: Block Diagram.kicad_sch

Revision History

File: Revision History.kicad_sch

Project Architecture

File: Project Architecture.kicad_sch

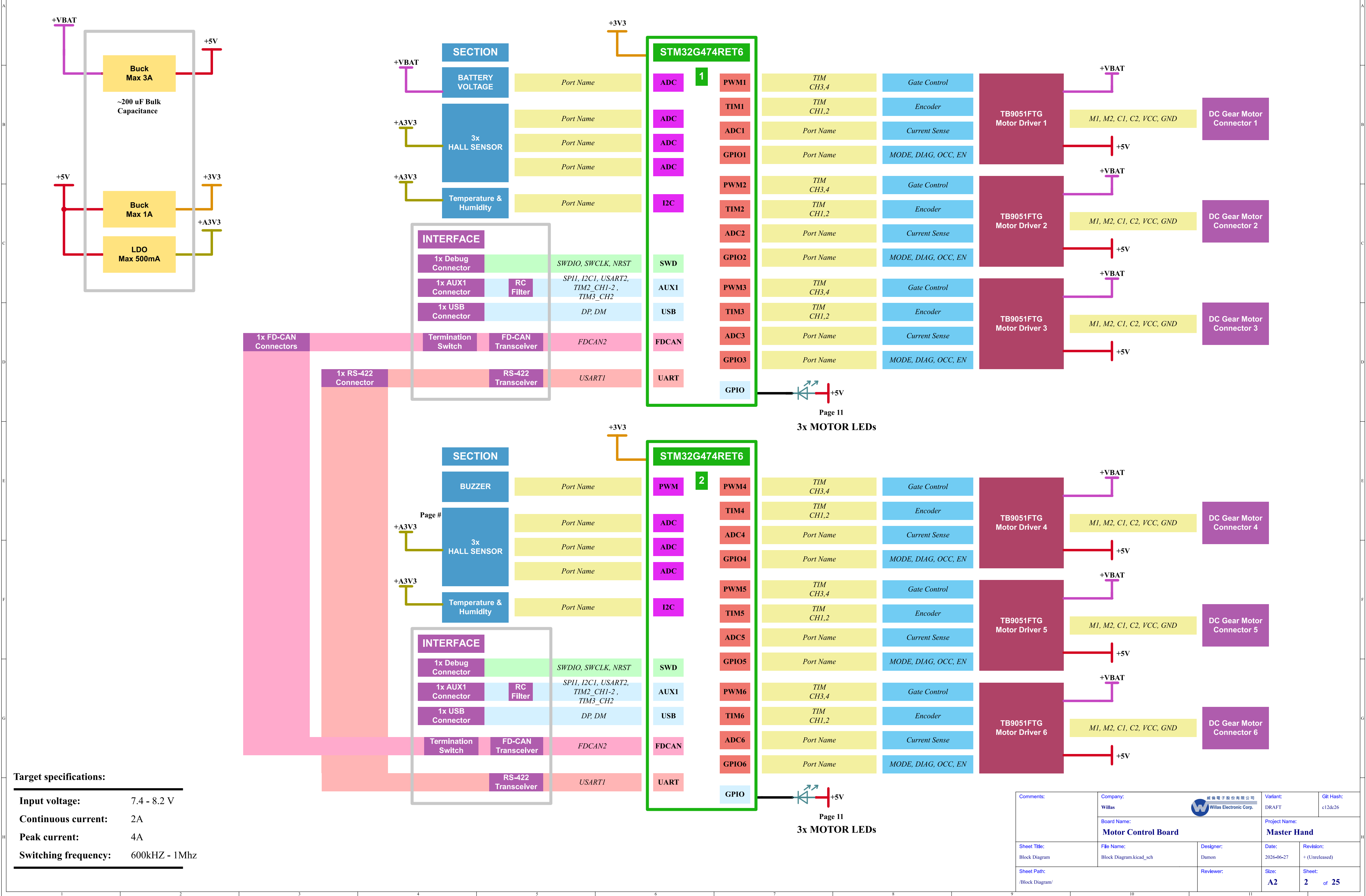
Power - Sequencing

File: Power - Sequencing.kicad_sch

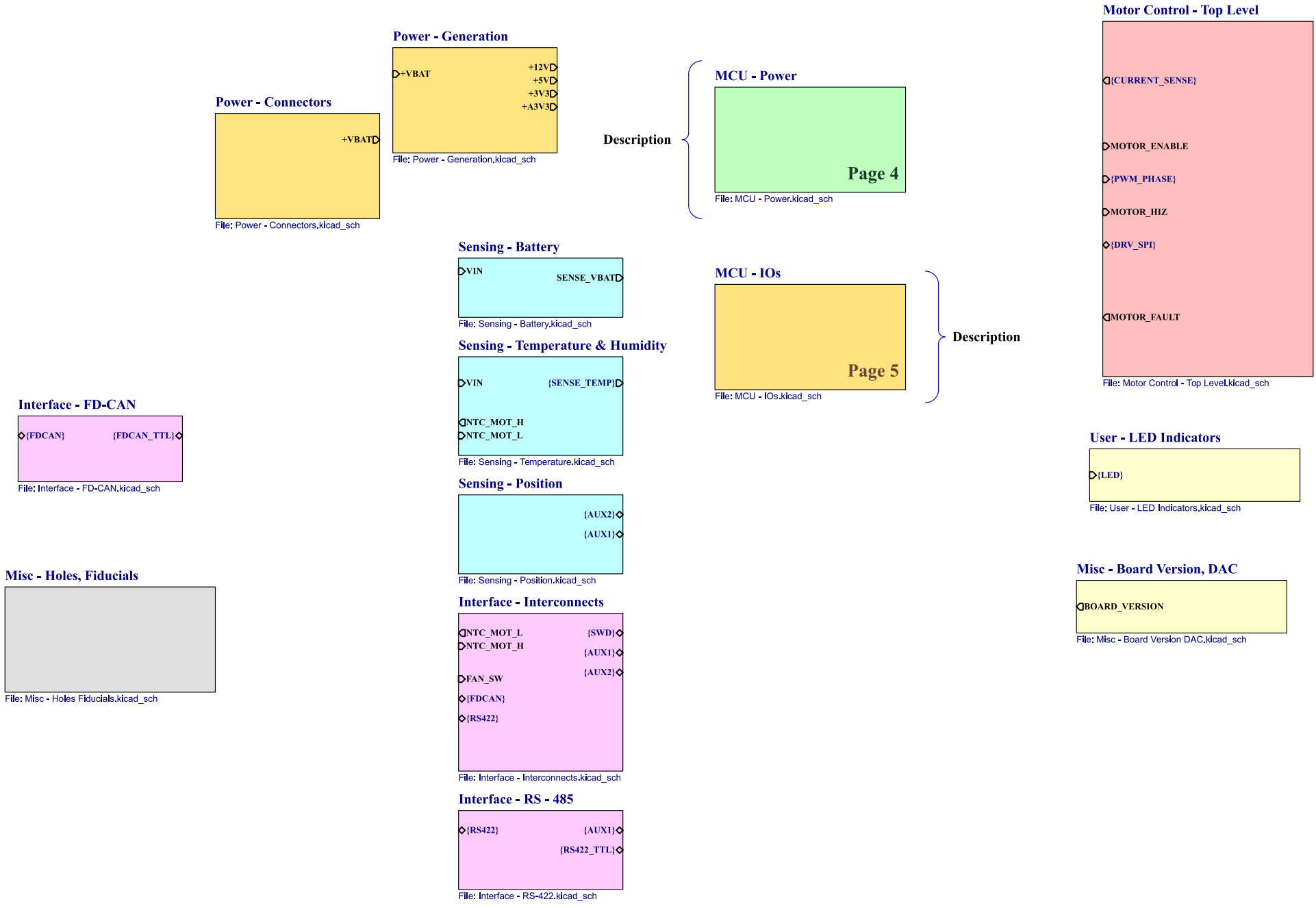
Date: 10-Aug-2026

Comments:	Company: Willas		W Willas Electronic Corp.		Variant: DRAFT	Git Hash: c12dc26
	Board Name: Motor Control Board				Project Name: Master Hand	
Sheet Title:	File Name: Master_Hand_Kicad.kicad_sch		Designer: Damon	Date: 2026-06-27	Revision: + (Unreleased)	
Sheet Path: /			Reviewer:	Size: A3	Sheet: 1 of 25	

[2] Block Diagram



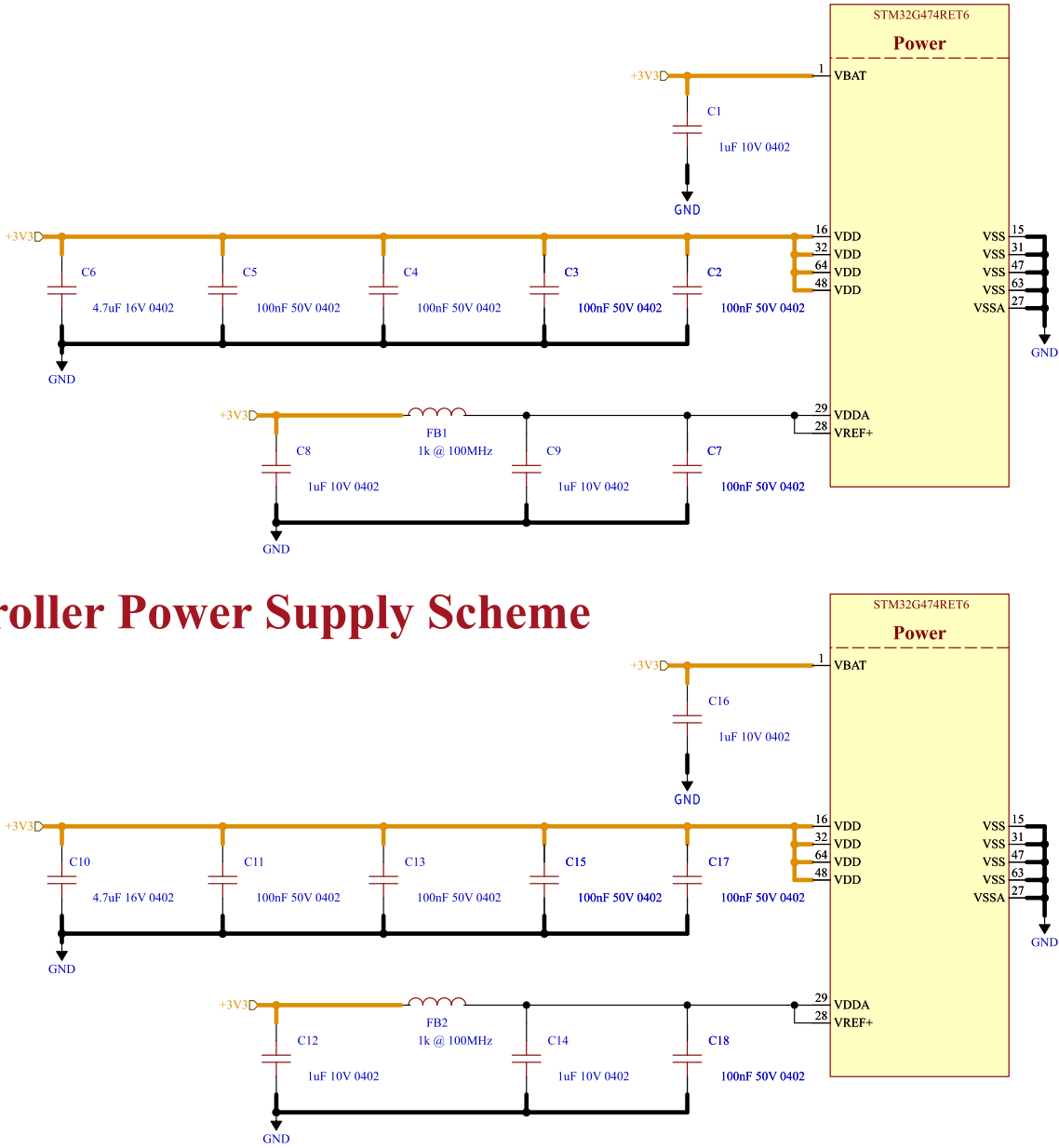
[3] Project Architecture



Comments:	Company: Willas  威倫電子股份有限公司 Willas Electronic Corp.		Variant: DRAFT		Git Hash: c12dc26
	Board Name: Motor Control Board		Project Name: Master Hand		
Sheet Title: Project Architecture	File Name: Project Architecture.kicad_sch	Designer: Damon	Date: 2026-06-27	Revision: + (Unreleased)	
Sheet Path: /Project Architecture/		Reviewer:	Size: A3	Sheet: 3 of 25	

[4] MCU - Power

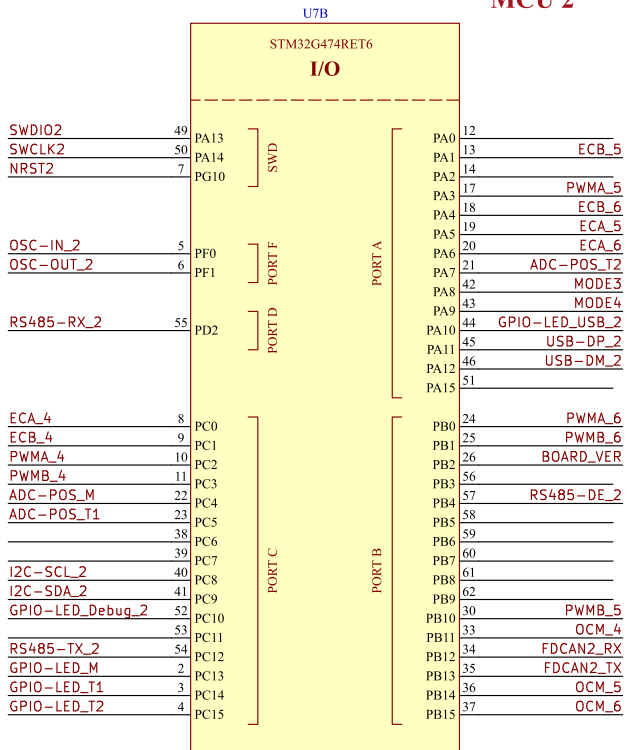
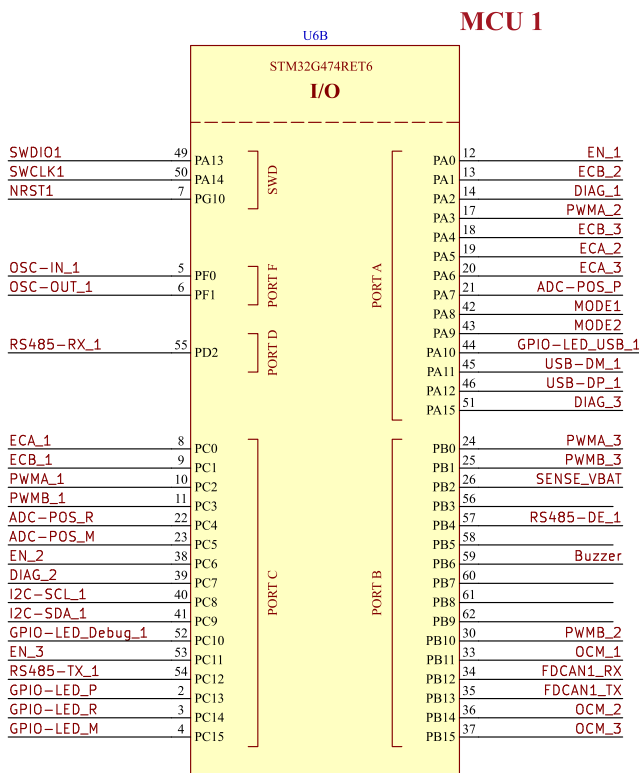
Microcontroller Power Supply Scheme



LAYOUT NOTE:
Place each capacitor close
to the relevant pin.
Minimize loop inductance.

Comments:	Company: Willas		威倫電子股份有限公司 Willas Electronic Corp. 		Variant: DRAFT	Git Hash: c12dc26
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Sheet Title: MCU - Power	File Name: MCU - Power.kicad_sch	Designer: Damon	Date: 2026-06-27	Revision: + (Unreleased)		
Sheet Path: /Project Architecture/MCU - Power/		Reviewer:	Size: A3	Sheet: 4 of 25		

[5] MCU - IOs



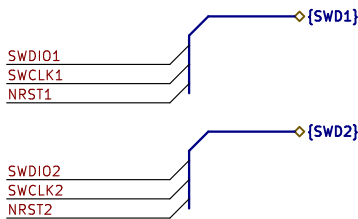
SENSE_VBAT SENSE_VBAT

Battery Voltage Sense

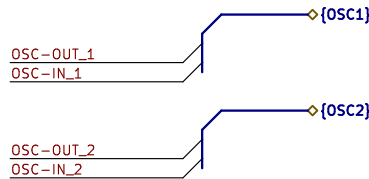
BOARD_VER BOARD_VERSION

Board Version Selection

Auxiliary Ports



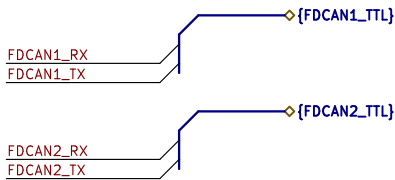
SWD (Debug) Port



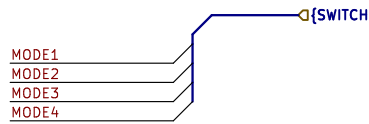
Oscillator

Buzzer Buzzer

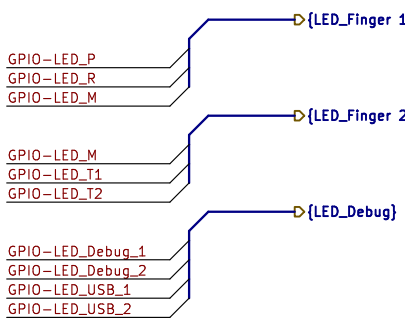
Buzzer



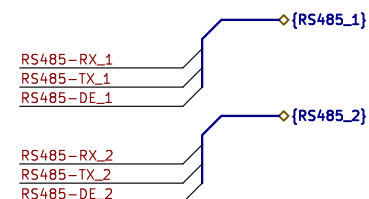
5Mbps FD-CAN Bus



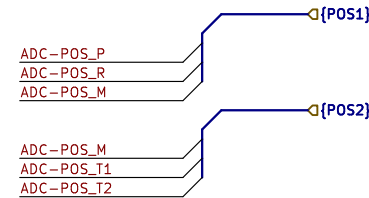
Switch



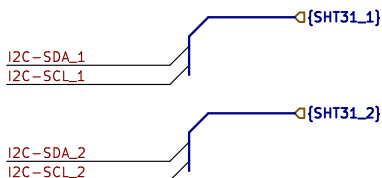
LEDs



RS - 485 Port



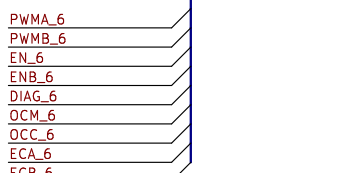
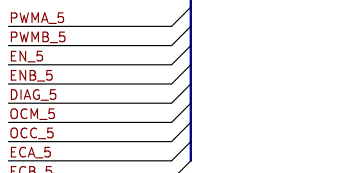
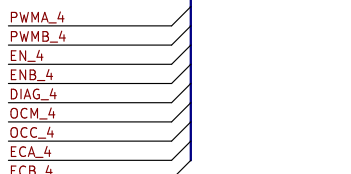
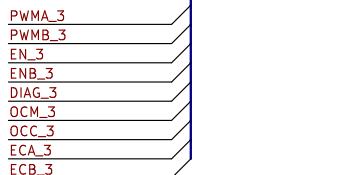
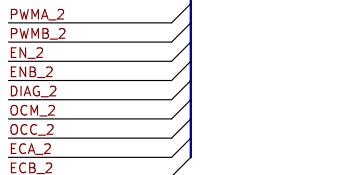
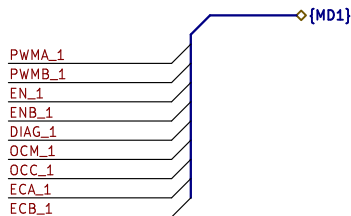
Hall Sensor - Position



Temperature and Humidity



USB PORT



Motor Driver

Motor Driver

PWMA_1 TP1
PWMB_1 TP2
DIAG_1 TP3
OCM_1 TP4

PWMA_2 TP5
PWMB_2 TP6
DIAG_2 TP7
OCM_2 TP8

PWMA_3 TP9
PWMB_3 TP10
DIAG_3 TP11
OCM_3 TP12

PWMA_4 TP13
PWMB_4 TP14
DIAG_4 TP15
OCM_4 TP16

PWMA_5 TP17
PWMB_5 TP18
DIAG_5 TP19
OCM_5 TP20

PWMA_6 TP21
PWMB_6 TP22
DIAG_6 TP23
OCM_6 TP24

Debug

SWDIO1 TP25
SWCLK1 TP26
NRST1 TP27

SWDIO2 TP28
SWCLK2 TP29
NRST2 TP30

FD-CAN

FDCAN1_RX TP31
FDCAN1_TX TP32
FDCAN2_RX TP33
FDCAN2_TX TP34

RS-485

RS485-RX_1 TP35
RS485-TX_1 TP36
RS485-DE_1 TP37

RS485-RX_2 TP38
RS485-TX_2 TP39
RS485-DE_2 TP40

Temperature & Humidity

I2C-SDA_1 TP41
I2C-SCL_1 TP42

I2C-SDA_2 TP43
I2C-SCL_2 TP44

Hall Position

ADC-POS_P TP45
ADC-POS_R TP46
ADC-POS_M TP47

ADC-POS_M TP48
ADC-POS_T1 TP49
ADC-POS_T2 TP50

Test Points

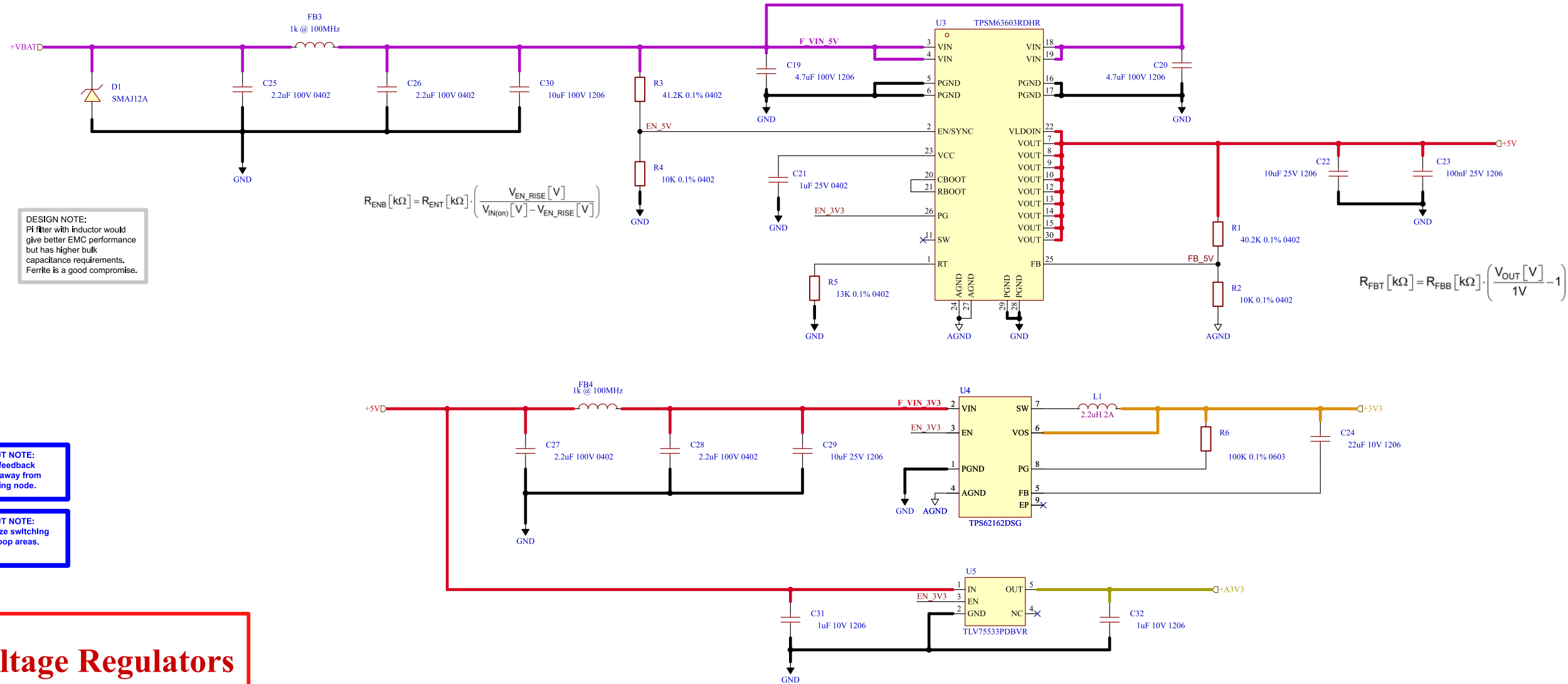
DESIGN NOTE:
We use the LQFP64
version to have extra
pins for controlling an
external fan.

DESIGN NOTE:
Do not forget to flash
option bytes when
flashing for the first
time.

LAYOUT NOTE:
Footprint deviates from
datasheet and IPC7351 to
allow for soldermask sliver
between pads according to
PCBWay design rules.

Comments:	Company: Willas Willas Electronic Corp.	Variant: DRAFT	Git Hash: c12dc26
Sheet Title: MCU - IOs	Board Name: Motor Control Board	Project Name: Master Hand	Revision: + (Unreleased)
Sheet Path: /Project Architecture/MCU - IOs/	File Name: MCU - IOs.kicad_sch	Designer: Damon	Size: A3
		Reviewer:	Sheet: 5 of 25

[6] Power - Generation



Voltage Regulators

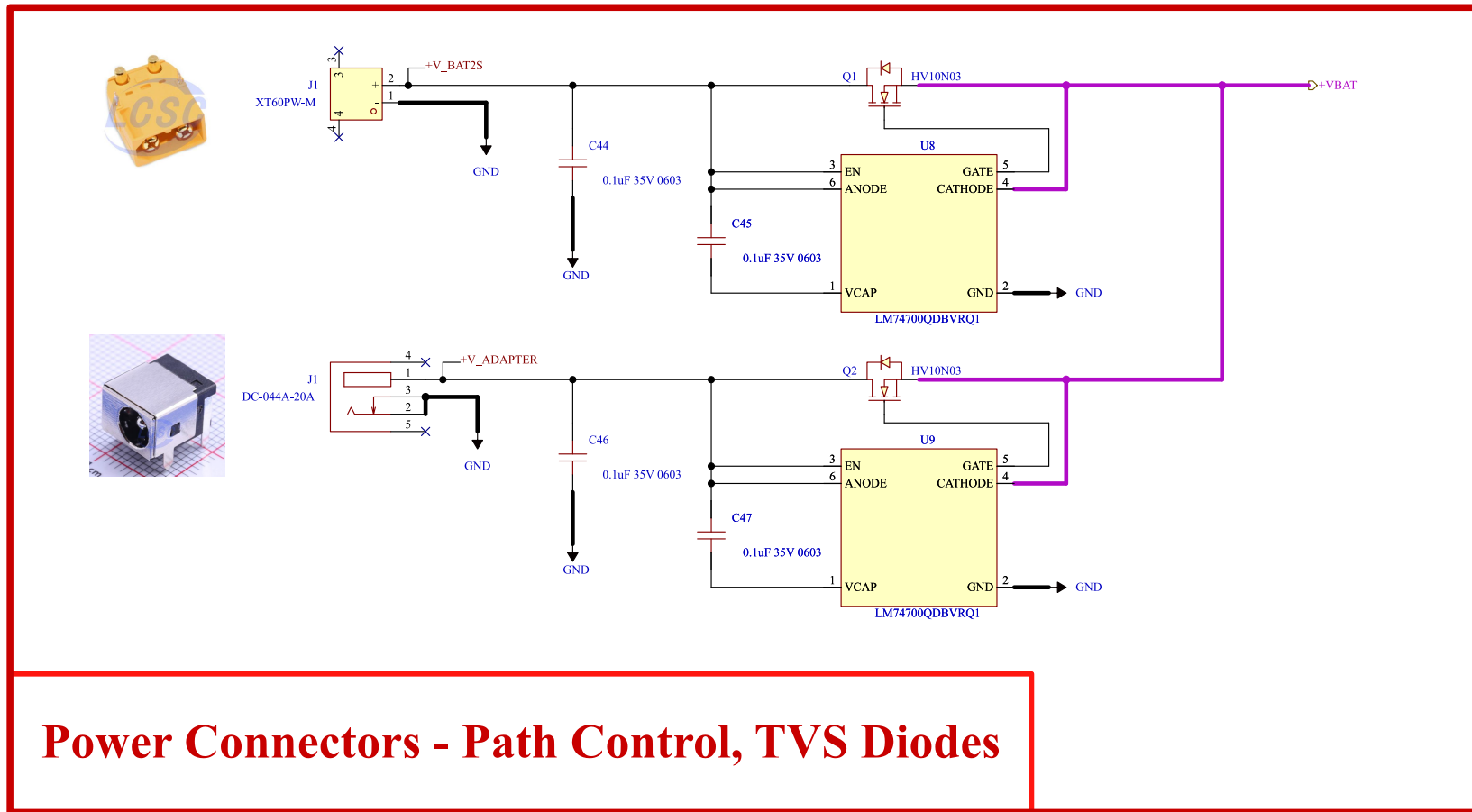
DC BULK

DC-Link Bulk Capacitors. 330uF + 10 x 10 uF = 430uF (35V)

Test Points

Comments:	Company: Willas  威倫電子股份有限公司 Willas Electronic Corp.		Variant: DRAFT		Git Hash: c12dc26
	Board Name: Motor Control Board		Project Name: Master Hand		
Sheet Title: Power - Generation	File Name: Power - Generation.kicad_sch	Designer: Damon	Date: 2026-06-27	Revision: + (Unreleased)	
Sheet Path: /Project Architecture/Power - Generation/		Reviewer:	Size: A3	Sheet: 6 of 25	

[7]



Power Connectors - Path Control, TVS Diodes

Comments:	Company: Willas 威倫電子股份有限公司 Willas Electronic Corp.		Variant: DRAFT	Git Hash: c12dc26
	Board Name: Motor Control Board		Project Name: Master Hand	
Sheet Title:	File Name: Power - Connectors.kicad_sch	Designer: Damon	Date: 2026-06-27	Revision: + (Unreleased)
Sheet Path: /Project Architecture/Power - Connectors/		Reviewer:	Size: A4	Sheet: 7 of 25



▷GATE_HS_A
 ◁SRC_HS_A
 ▷GATE_LS_A

- ◇SHUNT_P_A
- ◇SHUNT_N_A

▷GATE_HS_B
 ◁SRC_HS_B
 ▷GATE_LS_B

- ☐ SHUNT_P_B
- ☐ SHUNT_N_B

▷GATE_HS_C
◁SRC_HS_C
▷GATE_LS_C

- ◇SHUNT_P_C
- ◇SHUNT_N_C

File: Motor Control - Index.kicad sch



▷GATE_HS_A
 ◁SRC_HS_A
 ▷GATE_LS_A

SHUNT_P_A
SHUNT_N_A

▷GATE_HS_B
 ◁SRC_HS_B
 ▷GATE_LS_B

◇SHUNT_P_B
◇SHUNT_N_B

☒ GATE_HS_C
☐ SRC_HS_C
☒ GATE_LS_C

◇SHUNT_P_C
◇SHUNT_N_C

File: Motor Control - Ring.kicad sch



▷GATE_HS_A
 ◁SRC_HS_A
 ▷GATE_LS_A

SHUNT_P_A
SHUNT_N_A

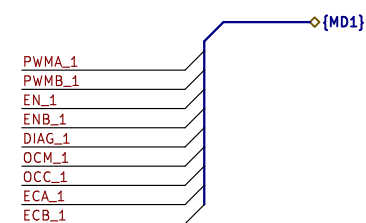
▷GATE_HS_B
 ◁SRC_HS_B
 ▷GATE_LS_B

SHUNT_P_B
SHUNT_N_B

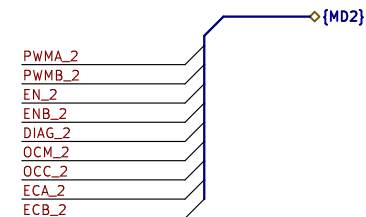
☒ GATE_HS_C
☐ SRC_HS_C
☒ GATE_LS_C

SHUNT_P_C
SHUNT_N_C

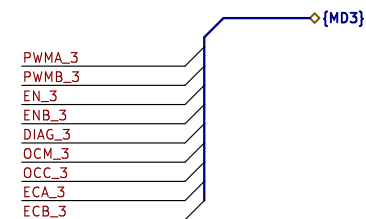
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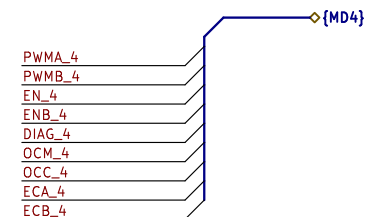
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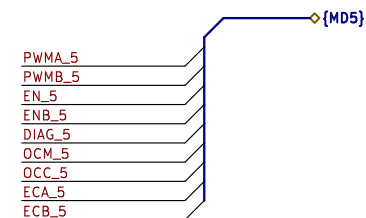
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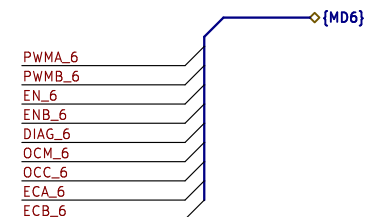
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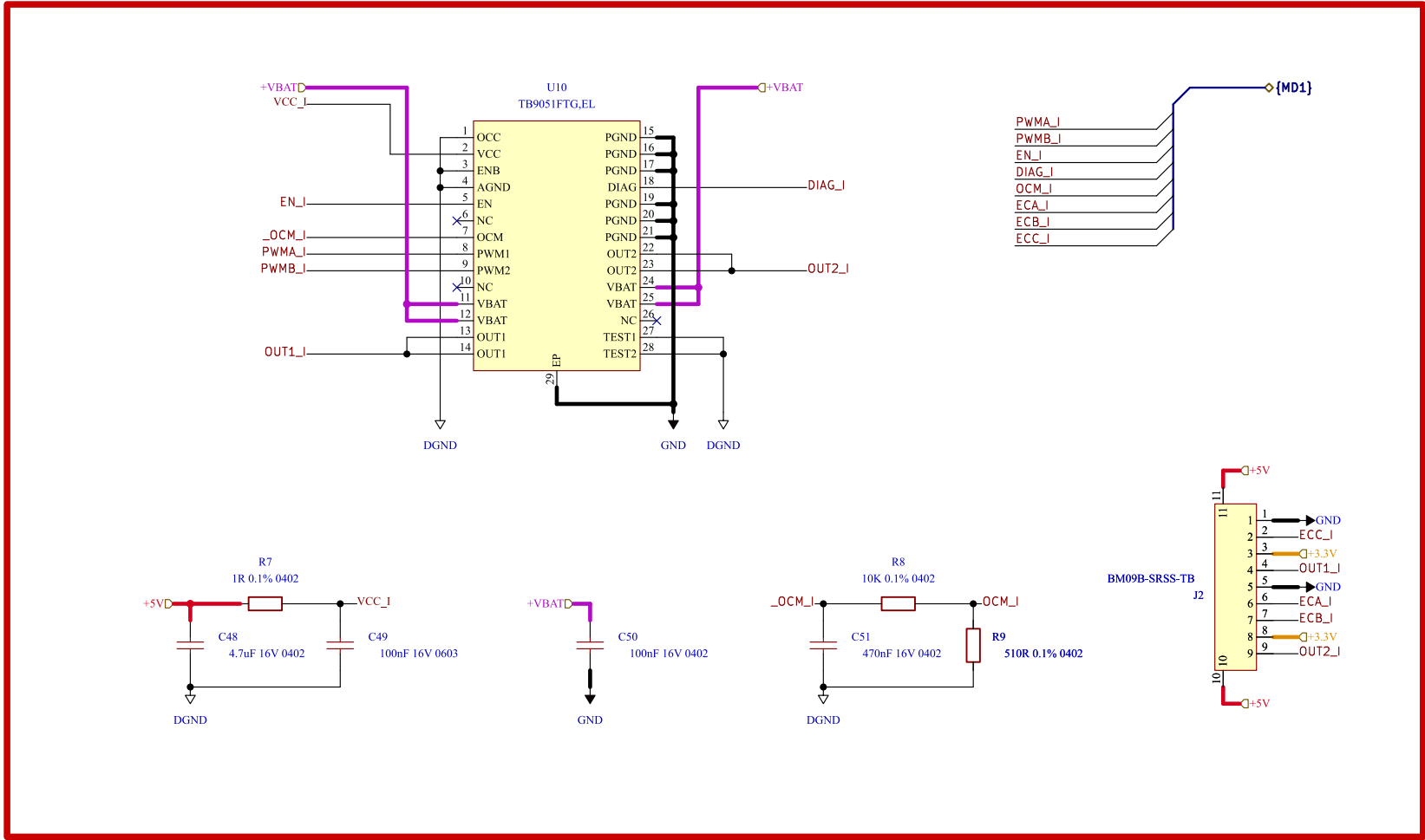


—◇ {MD5}



—◇ {MD6}

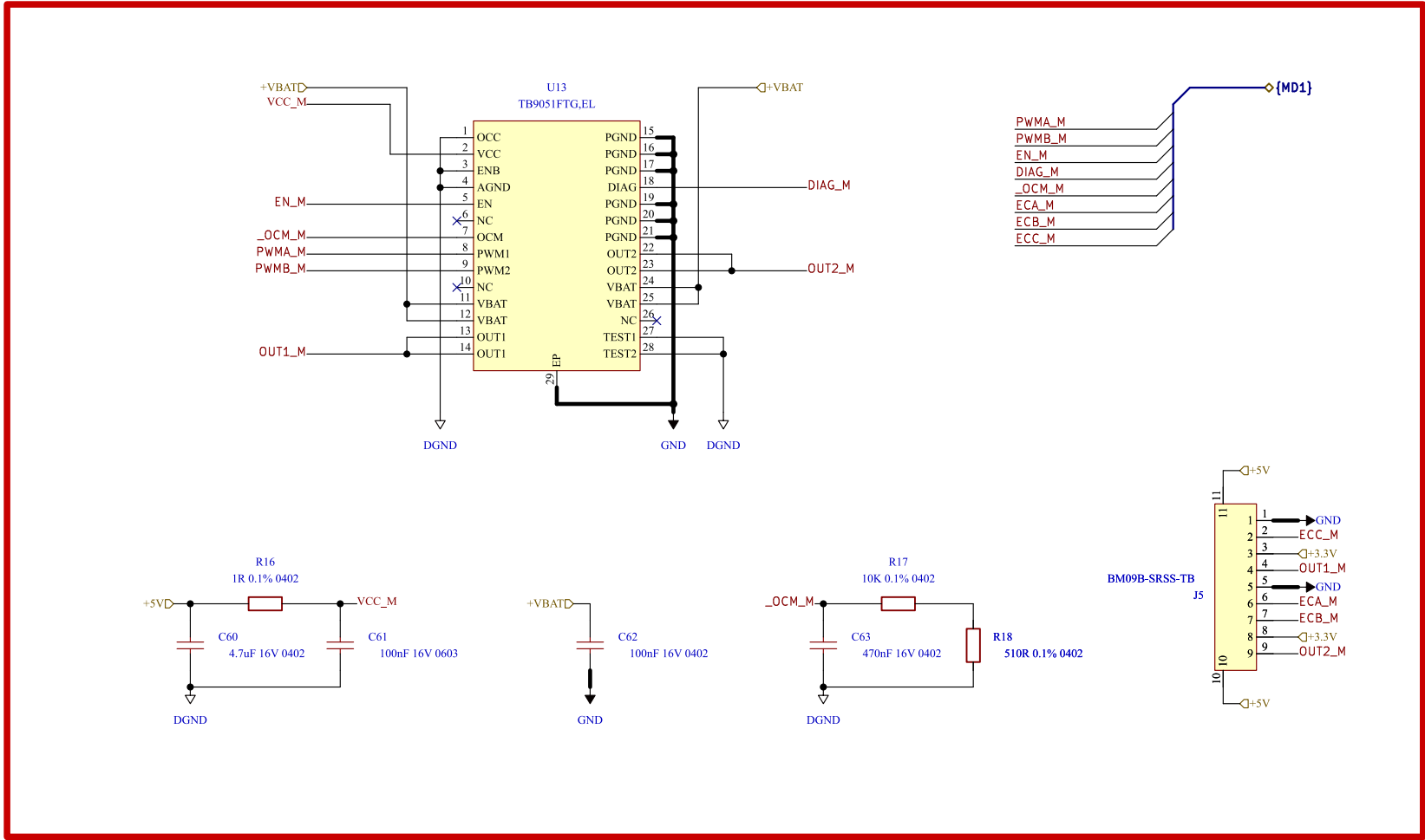
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	Board Name: Motor Control Board			Project Name: Master Hand		
Sheet Title:	File Name: Motor Control - Top Level.kicad_sch	Designer: Damon	Date: 2026-06-27	Revision: + (Unreleased)		
Sheet Path: //Project Architecture/Motor Control - Top Level/		Reviewer:	Size: A3	Sheet: 8 of 25		



Comments:	Company: Willas Willas Electronic Corp.		Variant: DRAFT	Git Hash: c12dc26
	Board Name: Motor Control Board		Project Name: Master Hand	
Sheet Title:	File Name: Motor Control - Index.kicad_sch	Designer: Damon	Date: 2026-06-27	Revision: + (Unreleased)
Sheet Path: /Project Architecture/Motor Control - Top Level/Motor Control - Index/		Reviewer:	Size: A4	Sheet: 18 of 25







Comments:

Company:

Willas



Variant:

DRAFT

Git Hash:

c12dc26

Board Name:

Motor Control Board

Project Name:

Master Hand

Sheet Title:

File Name:

Motor Control - Pinky.kicad_sch

Designer:

Damon

Date:

2026-06-27

Revision:

+(Unreleased)

Sheet Path:

/Project Architecture/Motor Control - Top Level/Motor Control - Pinky/

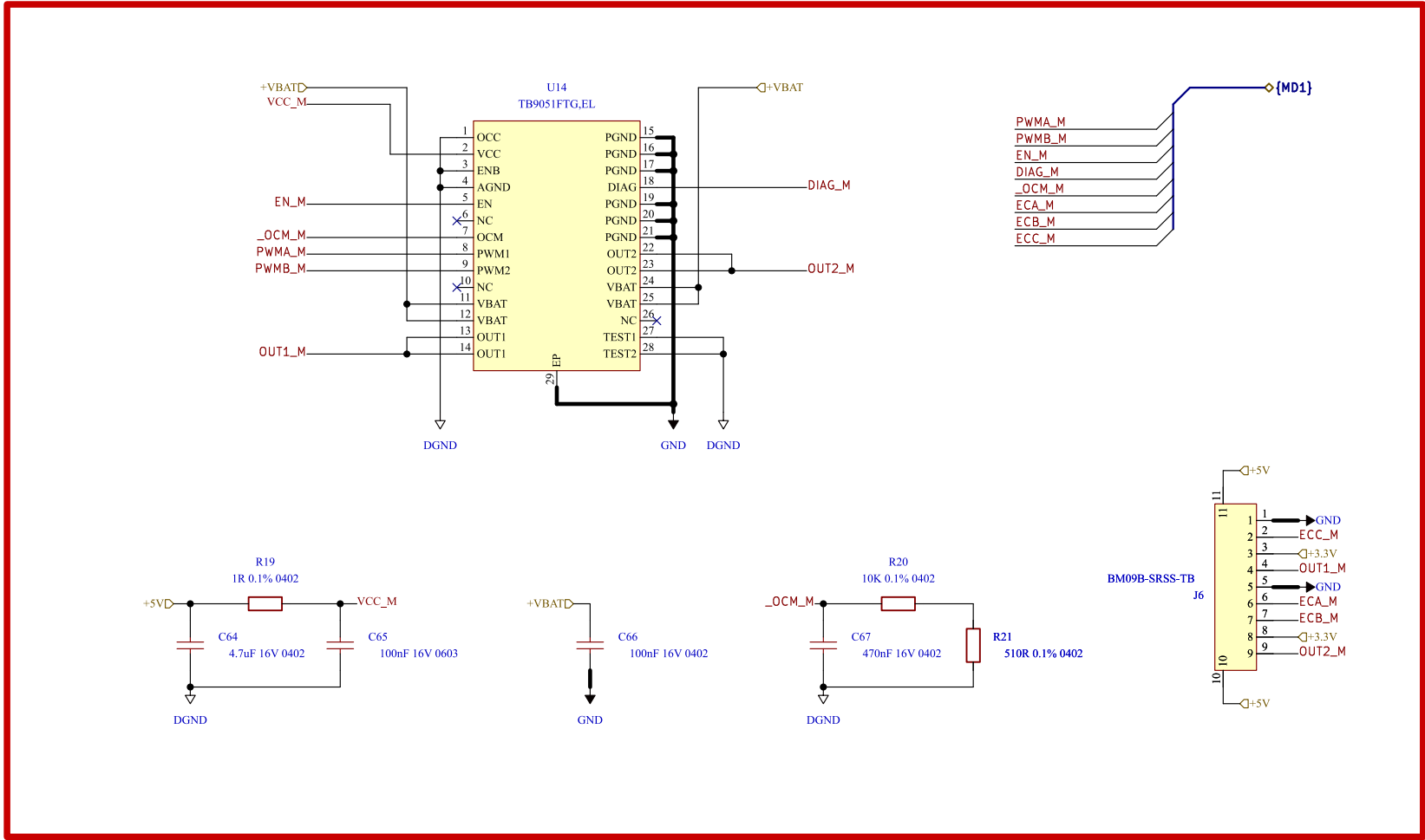
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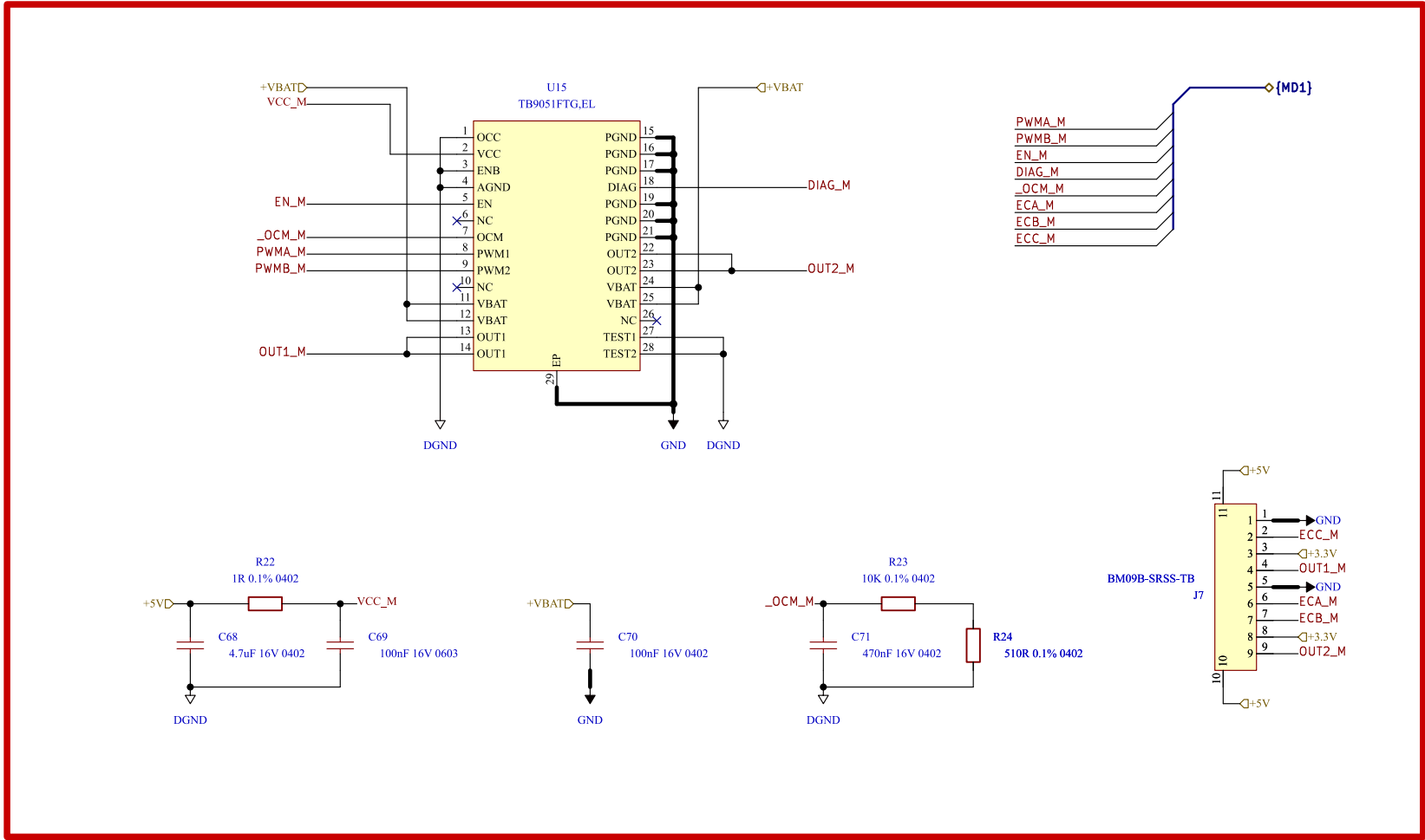
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Comments:	Company:  威倫電子股份有限公司 Willas Electronic Corp. Willas		Variant: DRAFT		Git Hash: c12dc26	
	Board Name: Motor Control Board			Project Name: Master Hand		
Sheet Title:	File Name: Motor Control - Thumb 2.kicad_sch	Designer: Damon	Date: 2026-06-27	Revision: + (Unreleased)		
Sheet Path: /Project Architecture/Motor Control - Top Level/Motor Control - Thumb 2/		Reviewer:	Size: A4	Sheet: 23 of 25		

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A																								
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D	<table><tr><td rowspan="2">Comments:</td><td colspan="2">Company: Willas </td><td>Variant: DRAFT</td><td>Git Hash: c12dc26</td></tr><tr><td colspan="2">Board Name: Motor Control Board</td><td colspan="2">Project Name: Master Hand</td></tr><tr><td>Sheet Title:</td><td>File Name: Misc - Board Version DAC.kicad_sch</td><td>Designer: Damon</td><td>Date: 2026-06-27</td><td>Revision: + (Unreleased)</td></tr><tr><td colspan="2">Sheet Path: /Project Architecture/Misc - Board Version, DAC/</td><td>Reviewer:</td><td>Size: A4</td><td>Sheet: 9 of 25</td></tr></table>					Comments:	Company: Willas 		Variant: DRAFT	Git Hash: c12dc26	Board Name: Motor Control Board		Project Name: Master Hand		Sheet Title:	File Name: Misc - Board Version DAC.kicad_sch	Designer: Damon	Date: 2026-06-27	Revision: + (Unreleased)	Sheet Path: /Project Architecture/Misc - Board Version, DAC/		Reviewer:	Size: A4	Sheet: 9 of 25
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B						B
C						C
D						D

Comments:	Company: Willas		Variant: DRAFT		Git Hash: c12dc26
	Board Name: Motor Control Board		Project Name: Master Hand		
Sheet Title:	File Name: User - LED Indicators.kicad_sch	Designer: Damon	Date: 2026-06-27	Revision: + (Unreleased)	
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B					B
C					C
D					D
1	2	3	4	5	6

Comments:	Company: Willas				Variant: DRAFT	Git Hash: c12dc26
	Board Name: Motor Control Board				Project Name: Master Hand	
Sheet Title:	File Name: Sensing - Battery.kicad_sch		Designer: Damon		Date: 2026-06-27	Revision: + (Unreleased)
Sheet Path: /Project Architecture/Sensing - Battery/			Reviewer:		Size: A4	Sheet: 12 of 25

1	2	3	4	5	6	
A						A
B						B
C						C
D						D

Comments:	Company: Willas		Variant: DRAFT		Git Hash: c12dc26
	Board Name: Motor Control Board		Project Name: Master Hand		
Sheet Title:	File Name: Sensing - Position.kicad_sch	Designer: Damon	Date: 2026-06-27	Revision: + (Unreleased)	
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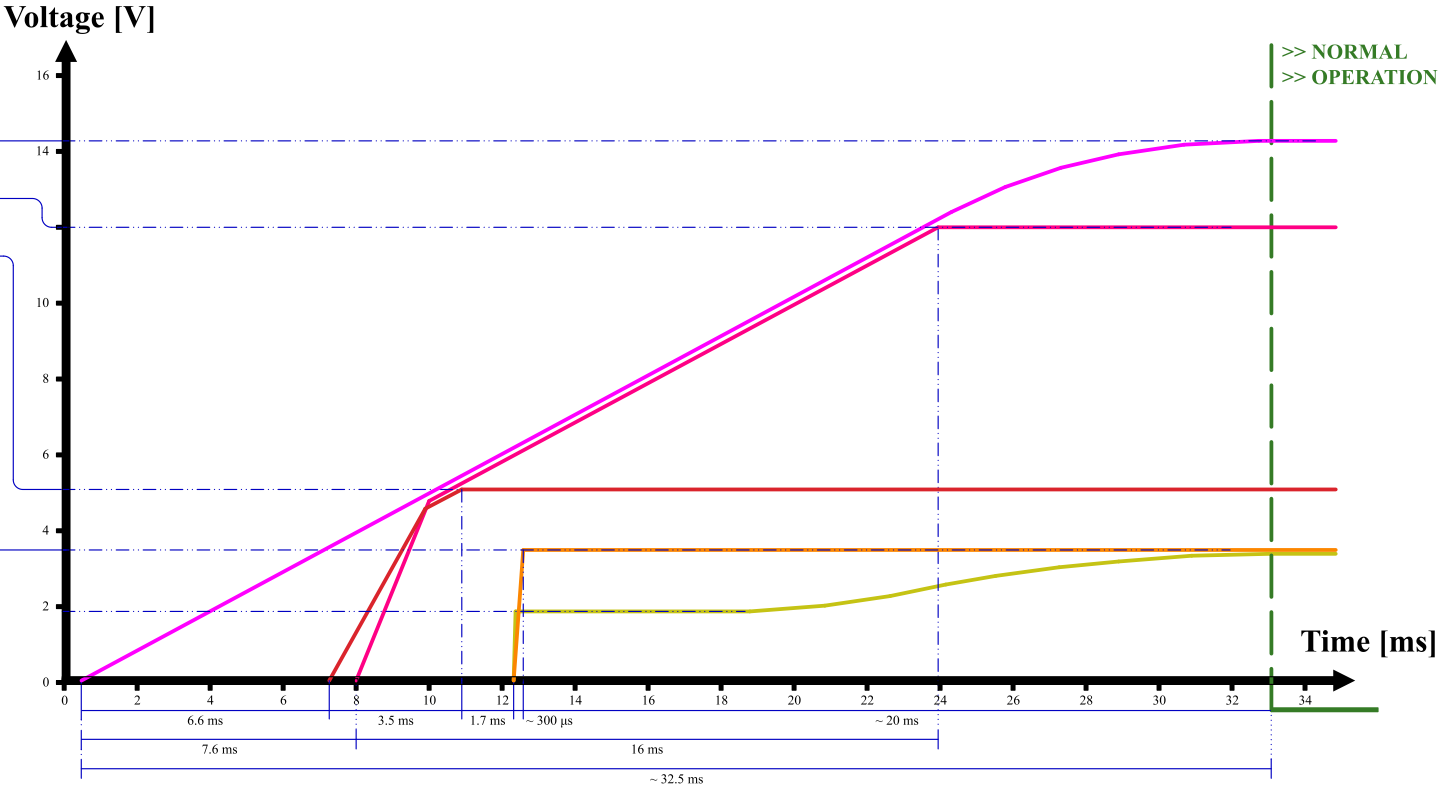
[illegible]

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D	<table><tr><td rowspan="2">Comments:</td><td colspan="2">Company: Willas </td><td>Variant: DRAFT</td><td colspan="2">Git Hash: c12dc26</td></tr><tr><td colspan="2">Board Name: Motor Control Board</td><td colspan="3">Project Name: Master Hand</td></tr><tr><td>Sheet Title:</td><td colspan="2">File Name: Interface - Interconnects.kicad_sch</td><td>Designer: Damon</td><td>Date: 2026-06-27</td><td colspan="2">Revision: + (Unreleased)</td></tr><tr><td colspan="3">Sheet Path: /Project Architecture/Interface - Interconnects/</td><td>Reviewer:</td><td>Size: A4</td><td colspan="2">Sheet: 16 of 25</td></tr></table>						Comments:	Company: Willas 		Variant: DRAFT	Git Hash: c12dc26		Board Name: Motor Control Board		Project Name: Master Hand			Sheet Title:	File Name: Interface - Interconnects.kicad_sch		Designer: Damon	Date: 2026-06-27	Revision: + (Unreleased)		Sheet Path: /Project Architecture/Interface - Interconnects/			Reviewer:	Size: A4	Sheet: 16 of 25	
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[6] Power - Sequencing

NAME	SOURCE	LEVEL
+V??	??	? - ? ± ??%
+V??	??	? - ? ± ??%
+V??	??	? - ? ± ??%
+V??	??	? - ? ± ??%
+V??	??	? - ? ± ??%



Comments:	Company: Willas Willas Electronic Corp.		Variant: DRAFT	Git Hash: c12dc26
	Board Name: Motor Control Board		Project Name: Master Hand	
Sheet Title: Power - Sequencing	File Name: Power - Sequencing.kicad_sch	Designer: Damon	Date: 2026-06-27	Revision: + (Unreleased)
Sheet Path: /Power - Sequencing/		Reviewer:	Size: A4	Sheet: 6 of 25

[7] Revision History

Comments:	Company: 威倫電子股份有限公司 Willas Willas Electronic Corp.		Variant: DRAFT		Git Hash: c12dc26
	Board Name: Motor Control Board		Project Name: Master Hand		
Sheet Title: Revision History	File Name: Revision History.kicad_sch	Designer: Damon	Date: 2026-06-27	Revision: + (Unreleased)	
Sheet Path: /Revision History/		Reviewer:	Size: A4	Sheet: 7 of 25	